

The consumer psychology of brands[☆]

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Abstract

This article presents a consumer-psychology model of brands that integrates empirical studies and individual constructs (such as brand categorization, brand affect, brand personality, brand symbolism and brand attachment, among others) into a comprehensive framework. The model distinguishes three levels of consumer engagement (object-centered, self-centered and social) and five processes (identifying, experiencing, integrating, signifying and connecting). Pertinent psychological constructs and empirical findings are presented for the constructs within each process. The article concludes with research ideas to test the model using both standard and consumer-neuroscience methods.

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Over the past two decades, we have learned a lot about the consumer judgments of brands and the processes that underlie specific brand-related phenomena, from brand extensions to global branding to brand equity. The empirical literature on brands is vast and detailed, demonstrating and testing highly domain-specific effects. But we have neglected investigating “the big picture”—identifying how specific empirical findings add up to a broader understanding of how consumers perceive brands. To be sure, domain-specific theorizing and testing is valuable and should continue. However, research on brands may benefit from a broader perspective that integrates various empirical findings into a comprehensive framework on the psychology of brands.

Consider the domain of brand extensions. The literature on brand extensions alone has amassed more than a hundred studies in leading journals and has identified numerous factors that affect how consumers feel toward a given brand extension. These factors have included: overall fit (Aaker & Keller, 1990; Bottomley & Holden, 2001; Milberg, Sinn, & Goodstein, 2010); type of brand (e.g., prestige or functional) (Monga & John, 2010); brand knowledge (Broniarczyk & Alba, 1994); the presence of explanatory links (Bridges, Keller, & Sood,

2000); parent-brand memory structures (Morrin, 1999); the strength of association between the brand’s parent category and the extension category (Herr, Farquhar, & Fazio, 1996); degree of congruence (Maoz & Tybout, 2002; Sood & Dreze, 2006); relatedness of the categories (Herr et al., 1996); sub-branding (Milberg, Park, & McCarthy, 1997); brand name suggestiveness (Sen, 1999); brand breadth (Sheinin & Schmitt, 1994); brand specific associations (Broniarczyk & Alba, 1994); brand extension typicality (Boush & Loken, 1991); intervening extensions (Keller & Aaker, 1992); positive affect (Adaval, 2003); brand attachment (Fedorikhin, Park, & Thomson, 2008); mood (Barone, Miniard, & Romeo, 2000); and involvement (Barone, 2005). What is missing from the literature is an analysis of how brand extension research contributes to our overall understanding of the consumer psychology of brands.

Not all of our research has been narrow and purely empirically-focused. We have also been quite inventive in generating new constructs—for example, brand personality (Aaker, 1997), brand relationships (Fournier, 1998), brand community (Muniz & O’Guinn, 2001), self-brand connections (Escalas, 2004), brand attachment (Thomson, MacInnis, & Park, 2005), and brand experience (Brakus, Schmitt, & Zarantonello, 2009).

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We have created scales to measure these constructs and others, such as brand trust (Delgado-B, Munuera-Aleman, & Yagoe-Guillin, 2003) and brand love (Carroll & Ahuvia, 2006). Yet, it is not clear how these constructs relate to each other and what specific role they play in the overall consumer psychology of brands.

Finally, several reviews and summaries of our research have been conducted. For example, Keller (2002) provided an exhaustive review of the literature on brands and brand equity. Keller and Lehman (2006) also reviewed the research and listed a large number of potential new research questions on brand positioning, brand personality, brand relationships, brand experience, corporate image and reputation, the integration of brand elements, channels and communications, company-controlled and external events, brand performance assessment and brand strategy issues (including brand extensions, brand architecture, co-branding, global branding, and branding and social welfare). Because we lack a general framework on the consumer psychology of brands, however, we do not know how answers to these empirical questions would enrich our understanding of brands significantly beyond what we know already.

My goal in this article is to move beyond domain-specific findings and individual brand constructs. I will identify the key brand constructs related to consumer psychology and integrate them into a comprehensive model. This consumer-psychology model of brands does not focus on brand outcomes, such as brand choice, purchase, or loyalty, but on the underlying psychological constructs and processes that contribute to such outcomes.

Comprehensive brand models and higher-level brand frameworks have been presented before in managerial articles, in textbooks and in trade books. Indeed, many important conceptual ideas were proposed first, or developed significantly, in such writings—for example, brand concept-image management (Park, Jaworski, & MacInnis, 1986), brand equity (Aaker, 1991; Keller, 1993); brand architecture and portfolio management (Aaker, 1996; 2004); and customer experience (Pine & Gilmore, 1999; Schmitt, 1999; 2003). Most of these models, however, do not take a consumer psychology angle. They present strategic typologies rather than conceptual frameworks rooted in consumer psychology. These models target marketing managers and not consumer psychologists, who I consider to be the prime audience of this journal and this article.

A consumer-psychology model of brands

The model presented here addresses consumer perceptions and judgments and their underlying processes *as they relate to brands*. Fig. 1 shows the model.

In contrast to general information processing models, the consumer-psychology model of brands focuses specifically on the unique characteristics of brands. One brand, for example, can span across various products and product categories. Brand information is conveyed frequently through multi-sensory stimulation. Brands can form relations with other brands. Brands can be anthropomorphized, and many of them are appreciated as cultural symbols. Finally, consumers can organize communities around brands. Consumers know and experience these characteristics

about brands and respond to them. The model presented here accounts for these essential characteristics of brands.

The structure of the model also reflects an understanding that consumers have different levels of psychological engagement with brands because of different needs, motives and goals. These levels of engagement are represented in the model by three layers. The innermost layer represents object-centered, functionally-driven engagement; that is, the consumer acquires information about the brand with the goal of receiving utilitarian benefits from the brand. The middle layer represents a self-centered engagement; the brand is seen as personally relevant to the consumer. Finally, the outer layer represents social engagement with the brand; the brand is viewed from an interpersonal and socio-cultural perspective, and provides a sense of community. As we move from the inner to the outer layer, the brand becomes increasingly meaningful to the consumer.

Most importantly, the model distinguishes five brand-related processes: identifying, experiencing, integrating, signaling and connecting with the brand. As part of *identifying*, a consumer identifies the brand and its category, forms associations, and compares the relations between brands. *Experiencing* refers to sensory, affective and participatory experiences that a consumer has with a brand. *Integrating* means combining brand information into an overall brand concept, personality and relationship with the brand. *Signifying* refers to using the brand as an informational cue, identity signal and cultural symbol. Finally, *connecting* with the brand includes forming an attitude toward the brand, becoming personally attached to it and connecting with the brand in a brand community. These processes are not necessarily one-directional and linear, in the way that information processing is presented from encoding to choice. As will be discussed in more detail at the end of this article, processes may occur in different orders. Moreover, while each construct is assumed to be conceptually distinct, a given construct may overlap, to some degree, with another construct, and different constructs may interact.

Let's look at the constructs within each process in more detail. What happens during the processes of identifying, experiencing, integrating, signifying and connecting?

Identifying

The process of identifying refers to searching for, being exposed to and collecting information about the brand, its category and related brands. Depending on a consumer's level of psychological engagement, the identification process concerns primarily categorization, associations with the brand, or inter-brand relations.

Brand categorization

When consumers engage with a brand in an object-centered way, they are mostly concerned with the brand, its product category and how the two are related. The primary task is linking a brand (its name and logo) to a product category, or, for corporate brands, industry category. Stimulus or memory-based categorization is a prerequisite for pursuing a brand-related goal (Alba, Hutchinson, & Lynch, 1991); that is, a consumer must

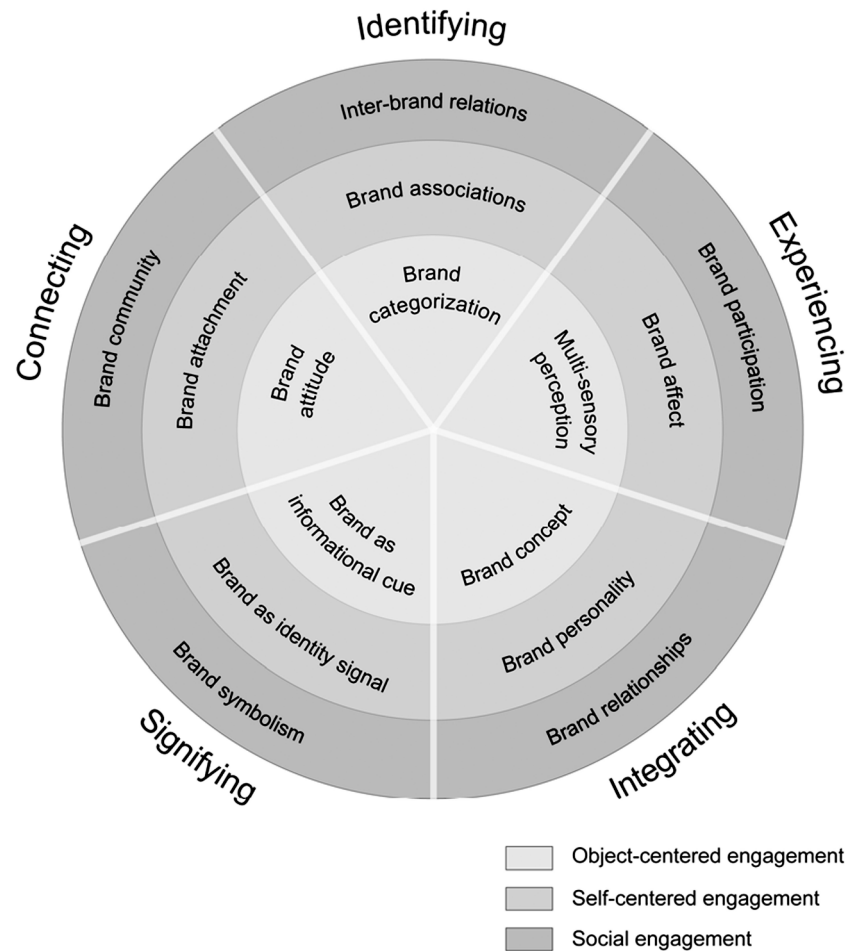


Fig. 1. Consumer psychology model of brands.

know at least the name and category to purchase a brand. At times, this may be enough: awareness of the link between a brand and its category may directly lead to choice (Hoyer & Brown, 1990; Nedungadi, 1990). Depending on the stimulus type, the categorization of a brand to a category may be established verbally (e.g., Kellogg's Corn Flakes) or visually through physical proximity (the brand name appears on the product packaging or web site), through temporal proximity (the name appears soon after a product shot in an ad), or through design (Kreuzbauer & Malter, 2005). There can be dilution effects, for example, when the distinction between brand names is blurred through similarity of the name or logo of a new brand to an existing brand (Pullig, Simmons, & Netemeyer, 2006).

Brand awareness is an important memory-based categorization task in which a consumer recalls a specific brand name when presented with the category. Memory depends on retrieval cues. Retrieval cues may be self-generated or externally-generated (Lynch & Srull, 1982). Two key retrieval cues that have been extensively studied are the product category and other brands (Alba & Chattopadhyay, 1985; Alba et al., 1991; Nedungadi, 1990). In addition, recall and recognition of a brand may be enhanced by linguistic characteristics and retrieval cues in communications and through lexical relations between the ad copy and

brand name (Keller, 1987; Lowrey, Shrum, & Dubitsky, 2003; Schmitt, Tavassoli, & Millard, 1993).

There is one additional key categorization task, namely identifying an additional category that may fit a brand, a topic that has been the subject of the extensive literature on brand extensions mentioned earlier. In brand extension research, consumers are typically asked about hypothetical new categories for existing brands. Consumers must thus decide whether a new product category should be integrated, and in the future be categorized, as belonging to an existing brand. Broadly speaking, two factors affect the relation between the brand and the extension: inter-category dynamics (such as overall fit, degree of congruence, intervening extensions) and brand specific dynamics (such as brand name suggestiveness, brand breadth, affect, attachment, and mood).

Brand associations

To engage with a brand in self-relevant ways, consumers identify information that is relevant to them. The unique set of brand associations that a brand strategist aspires to create or maintain in the consumer's mind constitutes a brand's identity (Aaker, 1996). This information may include, among

others, brand attributes, benefits, and images that the consumer encounters (Keller, 2003). Of course, in addition to associations provided by firms, consumers will develop their own associations with brands, for example, cognitive responses that the consumer generates about the brand (Keller, 2003). The information is stored as brand associations in consumer memory.

Research has employed associative network models, in which a person's memory is made up of links and nodes, to represent brand associations in the consumer's mind (Farquhar & Herr, 1993). Brand associations can differ in valence, strength, uniqueness and coherence (Keller, 1993). Brand associations are also structured in terms of level of generality. For example, some associations may be held about the overall brand (e.g., Sony) whereas others may focus on exemplars of the brand (e.g., Sony TVs) (Ng & Houston, 2006). The activation of brand associations is often automatic in nature. The so-called "mere association effect" can be detected through an implicit cognition measure (Dimofte & Yalch, 2011).

Inter-brand relations

Finally, to identify a brand on a social level, the consumer may be drawn to further information, namely, to the relations that the brand has with other brands. Brands become contextualized when brands are compared to other brands. Tell me who's your friend (and foe) and I'll tell you who you are.

Consumer responses to various forms of inter-brand relations have been studied: brand architecture and brand portfolios, including sub-branding and ingredient branding (Janiszewski & van Osselaer, 2000); co-branding (Geylani, Ter Hofstede, & Inman, 2005); and brand alliances (Simonin & Ruth, 1998). In each case, consumers compare a brand to another brand, which increases their understanding of the brands involved. Overall, one brand usually comes out as the winner in these inter-brand comparisons, by drawing greater consumer awareness or more positive associations. One of the most direct inter-brand relations is an explicit comparison, for example in the form of comparative advertising. In a meta-analysis, it was shown that comparative ads have a variety of benefits for brands although they reduce source credibility (Grewal, Kavanoor, Fern, Costley, & Barnes, 1997).

Experiencing

The experiencing process includes sensory perceptions of the brand, brand affect, and the participatory experiences that a consumer may seek with a brand. Research has conceptualized experiences as multi-dimensional, including sensory, affective-cognitive, and behavioral dimensions (Brakus et al., 2009).

Multi-sensory perception

At various contact points (or "touchpoints") with consumers, brands provide multi-sensory stimulations through sight, sound, smell, touch, and taste. When consumers are engaged with a brand in an object-centered, functional way, they pick up the multi-sensory stimuli of a brand (its logo, brand characters, verbal or auditory slogan) as presented in a store or on TV,

in print or banner ads rather mindlessly. At times, one sensory modality may dominate (think of paint, surround sound systems, fragrances, massage services, and ice cream brands). In perceiving most brands, however, more than one sense is involved: think of the roles that sight, sound, and touch play in evaluating a car brand. At consumption, many brands involve all five senses. Moreover, sensory cues within an environment can affect a brand; for instance, ambient scents can improve brand memory (Morris & Ratneshwar, 2003).

Although the human perceptual apparatus results in multi-sensory perceptions, most research in psychology and consumer psychology has been on the study of individual senses (Calvert, Spence, & Stein, 2004; Spence, 2010). Research has just begun to explore "cross-modal correspondences"—for example, how auditory cues while biting potato chips can affect the perception of crispiness or staleness of a potato chip (Zampini & Spence, 2004).

Psychophysicists have demonstrated that a person's memory for sensory attributes (e.g., intensity of light, depth of a color) decays very rapidly (Algom & Cain, 1991; Hubbard, 1994). However, when consumers are provided with a method to encode the sensory attribute meaningfully (e.g., Coca Cola Red, Tiffany Blue), memory for a sensory attributes improves drastically (Shapiro & Spence, 2002).

Finally, there are also implicit sensory effects in the verbal information presented about brands. Consumers can use sound symbolism to infer product attributes and evaluate brands (Yorkston & Menon, 2004). Also, as an example, pronouncing a brand name with a language accent can access cultural stereotypes and affects brand perceptions (Leclerc, Schmitt, & Dubé, 1994). Sensory processing may also occur implicitly for names because of the linguistic structure of a language. For instance, phonetically-based brand name translations from English to Chinese are evaluated differently than semantically based translations (Zhang & Schmitt, 2001). This effect seems to occur because reading logographs involves a higher degree of visual information processing than alphabetic English, which involves more phonetic processing (Tavassoli, 2001).

Brand affect

Beyond brands providing mere sensory stimulations, they may also evoke positive or negative moods and make consumers feel joyful and happy or angry and sad, especially when consumers engage with brands in a self-centered way. These positive and negative feelings can range from mild affect (e.g., positive moods) to strong affect (e.g., specific emotions). Indeed, brand research has examined a wide range of brand affect, from general measures of pleasure and feeling good to brand love (Carroll & Ahuvia, 2006; Chaudhuri & Holbrook, 2001). In brand management and advertising, so-called "emotional branding" has emerged as an alternative to earlier marketing approaches that focused purely on unique selling propositions (Gobé, 2001).

Emotions such as joy, sorrow, love, or anger are strong and object-directed and usually the outcome of an appraisal process (Smith & Ellsworth, 1985). Moods, in contrast are milder affective states, less focused and more diffused. Nonetheless moods can convey information about the brand when consumers use

the “how-do-I-feel-about-it?” heuristic (Pham, 2004; Schwarz, 1990). Consumers may also have ambiguous feelings about brands, e.g., enjoying a fast food brand or luxury handbag brand, yet feeling regret about consuming a lot of calories or paying a high price (Ramanathan & Williams, 2007). Finally, individuals can even feel positive and negative affect at the same time, e.g., fear and fun during a horror movie (Andrade & Cohen, 2007).

Brand participation

When consumers are socially engaged, they may experience the brand by actively participating and interacting with it. The model refers to such experiences as *brand participation*. That is, the consumer is no longer a passive recipient of information, or even an active processor of information stored in the mind. Rather, experiencing and doing are intertwined as part of a behavioral experience (Brakus et al., 2009).

Brand participation often occurs in conventional brand settings. The interactive atmosphere of retail environments, the immersive stimulation occurring at live events and the ability to customize product features also offer opportunities for participatory and interactive experiences (Akaoui, 2007). Some rapidly developing new media, referred to as “social media,” also allow for brand participation through digital sharing and “mashing-up” of brand related information. Brand participation seems to be a key contributor to the effectiveness of social media (Hoffman & Fodor, 2010). Moreover, brands like Starbucks have developed online crowd sourcing platforms where consumers can contribute creative ideas for new products, services and experiences (Bayus, 2010).

Integrating

During the integration process, consumers combine brand information and summarize it in an overall brand concept, personality or relationship with the brand.

Brand concept

Brand concept is a psychological construct consisting of the integrated information associated with a product brand or corporate brand. Brand concepts facilitate functionally-driven goal pursuit. The integrated information is usually stored in the form of a superordinate concept (e.g., as a “quality,” “innovative,” or “lifestyle” brand). The overall brand concept (or “image,” or “core”) has been considered an integral component of brand equity and has been widely employed in management-focused writings (Aaker, 1996; Agarwal & Rao, 1996; Keller, 1993; Park & Srinivasan, 1994; Park et al., 1986).

The information integration that results in a brand concept may be the outcome of some sort of cognitive algebra that, following Anderson’s information integration theory, weights the brand-related information acquired and stored in memory (Anderson, 1981). Also, certain information, particularly the information resulting from multi-sensory perceptions, may be more or less salient which affects its incorporation into the brand concept.

Finally, the information may be overall more or less well integrated. One methodology to elicit brand concepts from consumers is to create “Brand Concept Maps” (Roedder John, Loken, Kim, & Monga, 2006). The methodology allows researchers to determine how important brand associations are, whether they are direct or indirect associations, and how interconnected they are within a brand concept.

Brand personality

When consumers are engaged in a self-relevant way, information and experiences may be integrated further by inferring trait and personality characteristics about a brand (Aaker, 1997). By ascribing human characteristics to a brand, consumers are anthropomorphizing it (Aggarwal & McGill, 2007). These brand personalities are relatively stable over time but can vary in different consumption settings, in line with the idea of a “malleable self” (Aaker, 1999; Graeff, 1997). Most importantly, these inferred personalities differentiate brands in the mind of consumers even when consumers cannot articulate differences in associated attributes and benefits, or when there are few sensory differences. The colorless, odorless and tasteless vodka product category is a case in point. One vodka may be seen as “cool” and “hip,” whereas another may be described as “intellectual” and “conservative” (Aaker, 1997).

A five-factor structure—Sincerity, Excitement, Competence, Sophistication, and Ruggedness—seems to best display American consumers’ brand personality perceptions (Aaker, 1997). Three dimensions (Sincerity, Excitement, and Competence) resemble closely three human personality dimensions (Agreeableness, Extraversion, and Conscientiousness), whereas two dimensions (Sophistication and Ruggedness) are not consistent with those of the big-five human personality models (McCrae & Costa, 1997). However, the brand personality structure may not be universal (Caprara, Barbaranelli, & Guido, 2001). Only three of the five factors applied to brands in Japan and Spain (Aaker, Benet-Martínez, & Garolera, 2001). Peacefulness replaced Ruggedness both in Japan and Spain; Passion emerged in Spain instead of Competency. A revised brand personality scale exhibits cross-cultural validity between the U.S. and European markets (Geuens, Weijters, & De Wulf, 2009).

Brand relationships

In addition to assigning human-like properties to brands, consumers may also interact with brands in ways that parallel interpersonal and social relationships. Indeed, in qualitative research, Fournier (1998) found evidence for all sorts of customer-brand relationships, fifteen in total, including: arranged marriages, casual friends/buddies, marriages of convenience, committed partnerships, best friendships, compartmentalized friendships, kinships, rebounds/avoidance-driven relationships, childhood friendships, courtships, dependencies, flings, enmities, secret affairs, and enslavements.

In brand relationships, the norms of social relationships and social rules that govern society are used as guiding principles in brand interactions. When a brand's action violates the relationship norms, consumers assess the brand more negatively than when the brand's actions are consistent with those relationship norms (Aggarwal, 2004). Moreover, norms of a communal relationship, relative to those of an exchange relationship, make individuals more likely to process brand information at a higher level of abstraction (Aggarwal and Law, 2005).

Finally, brand personalities and relationships may interact. For example, relationships with "sincere" brands were found to deepen in line with "friendship" templates, whereas relationships with "exciting" brands displayed characteristics of "flings" (Aaker, Fournier, & Brasel, 2004).

Signifying

Semiotically, brands may be viewed as signifiers that transfer meaning (Mick, 1986). Depending on the consumer's engagement, a brand may act as an informational cue, personal identity signal or cultural symbol. Signifying may occur heuristically, without the need for extensive processing (Maheswaran, Mackie, & Chaiken, 1992).

Brands as informational cues

The accumulated information and knowledge about a brand can be used in a functional–rational way as informational cues. Price and quality of a brand are the most widely used types of informational signals, signifying that a brand is a value, premium or luxury brand (Zeithaml, 1988).

In a competitive marketplace, brands can be used by firms to inform consumers about product positions in the marketplace (Erdem & Swait, 1998). Brands can do so especially well when the signal that they convey is clear and consistent, and, most importantly, credible (that is, truthful and dependable).

Brands as identity signals

When the self is engaged, then the brand can be a signal for a consumer's personal identity, to both the consumer and to others. Psychological research has shown that the self consists of stable knowledge structures (so-called "self-schemas") that organize incoming self-related information and help people make sense of themselves in their environments (Markus, 1977). People vary in their tendency to possess particular self-schemas, and this variation leads to differential attitudes and behaviors toward objects, including brands (Markus, 1983; Markus & Sentis, 1982). Consumers with a strong masculine self-schema described fragrance brands in more accentuated gendered terms and held sharply different brand preferences than those with weaker masculine self-schemas (Markus & Sentis, 1982).

The part of the self that is defined by brands is referred to as self-brand identity. The self-brand connection scale measures the strength of the link between the self and a particular brand (Escalas, 2004). There are also individual differences among

consumers with respect to their tendency to include brands in their self-schemas (Sprott, Czelebar, & Spangenberg, 2009).

Consumers can use brands to express and display, and better understand, their selves (Swaminathan, Page, & Gürhan-Canli, 2007). That is, a brand may be used to express the self because the brand is congruent with the self ("I buy a Jeep because I view myself as tough"). A brand may also be used to display the self to others ("I buy a Jeep to show to others that I am tough") (Kirmani, 2009). Finally, consumers may infer their self and identity based on their own brand purchases ("I am tough because I bought a Jeep"), or, through a vicarious self-perception process, based on the purchases of others with whom they feel a merged identity (Goldstein & Cialdini, 2007).

Examining informational and identity signifiers in a developmental study, it was found that the eight-year-olds in the study treated brands as informational and perceptual cues: they associated a brand with a product category or certain perceptual features. In comparison, older children (starting at the age of twelve) understood the identity-shaping and symbolic qualities of a brand in terms of conveying meaning to the self, status, prestige, or trendiness (Chaplin & Roedder John, 2005).

Brand symbolism

Brands may be used to signify not only individual selves; they may also be used to represent a group, a society, or culture. As cultural symbols, they can stand for nations (McDonald's), generations (the Gap), and cultural values (Marlboro, Harley-Davidson). As exemplary symbols that are worthy of admiration and respect; they can assume the role of cultural icons and assume mythic qualities (Holt, 2004).

Consumer culture theory has illuminated the socio-cultural symbolism and ideology of brands (Arnould, 2005). Sociologically speaking, brands are social representations (Moscovici & Markova, 1998). They allow groups and communities to communicate, behave and orient themselves. Consumers may also use brands to enact archetype myths, e.g., the archetype of the hero, rebel or lover (Woodside, Sood, & Miller, 2008). Following Jungian analysis, archetypes are universally understood, recurring symbols in the collective unconscious that appear in cultural myths, literature and art, and, in contemporary consumer culture, as brand stories. Finally, brands can provide symbolic security, for example, for materialistic consumers who experience existential insecurities, e.g., fear of death (Rindfleisch, Burroughs, & Wong, 2009).

Connecting

Finally, the model distinguishes three psychological constructs to indicate various ways of connecting with a brand that differ in strength and affect the consumer's interaction with a brand: brand attitude (resulting from object-centered engagements with brands), brand attachment (resulting from self-centered engagements), and brand community (resulting from interpersonal and socio-cultural engagements).

Brand attitude

Brand attitudes are psychological tendencies to evaluate objects along a degree of favor or liking. Attitudes toward brands, or ads, have been central constructs in consumer psychology for a long time (MacKenzie, Lutz, & Belch, 1986; Mitchell & Olson, 1981). Recently, following dual-processing theories in psychology, a distinction has been drawn between implicit and explicit attitudes (Gawronski & Bodenhausen, 2006). The basis of implicit attitudes is seen in associative processes that are activated automatically with little cognitive capacity or explicit intention to evaluate an object. For brands, they may be the result of a classical conditioning process, e.g., by pairing sensory images with brands (Grossman & Till, 1998). Explicit attitudes, in contrast, are evaluative judgments that are derived through a reflective system and the resulting propositions are subject to syllogistic inferences that assess their validity.

Positive attitudes express a relatively weak connection with a brand. They are generalized dispositions to behave toward a brand, and they may lead to simple preference and purchase intention. But attitudes are often not stable over time, and the attitude–behavior link is weak and subject to numerous moderator effects (Park & MacInnis, 2006).

Brand attachment

For self-related engagement, brand attachment seems to be the essential construct that expresses a consumer's connection with a brand. Brand attachment provides stronger connections than brand attitudes (Thomson et al., 2005). Attachment was originally used in developmental psychology in the realm of parent–infant relationships to define a strong bond between a child and caretaker (Bowlby, 1979). After childhood, attachment can manifest itself in romantic relationships (Hazan & Shaver, 1994), kinships, and friendships (Trinke & Bartholomew, 1997; Weiss, 1988).

In the realm of consumer psychology, consumers can form emotional attachments to gifts, collectibles, places of residence, and, in particular, brands (Thomson et al., 2005). Brand attachment and brand attitude have distinct conceptual properties and formation processes, and, therefore, different behavioral implications (Park, MacInnis, Priester, Eisingerich, & Iacobucci, 2010). Brand attachment predicts consumer intentions to perform behaviors that use significant resources, such as time, money, and reputation, better than brand attitudes. Attachment may be viewed as an antecedent of true loyalty (Fournier & Yao, 1997).

Brand community

Moving to the interpersonal and socio-cultural engagement level, researchers have examined consumer connections with brand communities. A brand community is “a specialized, non-geographically bound community, based on a structured set of social relationships among users of a brand” (Muniz & O'Guinn, 2001). Brand communities are explicitly commercial and, unlike other communities (such as neighborhood groups, collectors or social clubs), not necessarily tied to geography.

They tend to form around publicly consumed goods rather than those consumed in private.

Brand community includes a sense of emotional involvement and connection with the group. Yet, brand communities are not only providing emotional bonds. They create shared goals among members, who may engage in joint actions to accomplish these collective goals (Bagozzi & Dholakia, 2006). They provide help and support, recommendations, and interaction with like-minded consumers, as well as participation in community activities. The meaningfulness of the community rests in sharing such activities and experiences (McAlexander, Schouten, & Koenig, 2002). As social entities, brand communities also have their own norms and rituals, and consumers tend to agree with the community's objectives, norms, and rituals (Algesheimer, Dholakia, & Herrmann, 2005).

Future research

The consumer-psychology model of brands presented here provides an integrative framework for organizing research on brands and guiding further empirical studies. Future research should test the validity of the core constructs and core assumptions of the model, regarding the three layers of engagement, the five brand related processes, and, most importantly, regarding the allocation of constructs to these layers and processes.

As mentioned earlier, the model proposes that the five psychological brand processes do not necessarily occur in a particular order. Although I proceeded didactically from identifying relevant information and experiencing the brand, to integrating information and experiences, to signifying and connecting with the brand, some of these processes may occur in alternative orders. For example, after trial, a consumer may have formed a positive attitude (e.g., toward a consumer electronics brand) that he only subsequently confirms through a brand concept (e.g., as an “innovative brand”). Or the consumer may endow a brand with a personality (“exciting”) and subsequently search for associated features and benefits that confirm it. Or, after becoming part of a brand community, a consumer may judge her brand relationship to be a “friendship.” Future research should investigate under what conditions the processes of the model occur sequentially from identifying to connecting and when they occur in a different sequence.

There should also be tests concerning whether the three layers of engagement are psychologically distinct. Is there, as proposed, a relatively basic object-centered and functionally-guided consumer engagement process that includes simple categorization and multi-sensory processing that can be summarized in a brand concept that serves as an informational cue for brand attitudes? Is there a self-centered engagement process that includes brand associations, affect and personality and serves as an identity signal that leads to attachment toward the brand? Finally, can we verify an interpersonal and social engagement process where consumers examine brands contextually and process inter-brand relations, create their own experiences through participation, relate to brands as cultural symbols and view themselves as part of a brand community?

In testing the model or parts of it, we should allow for moderator effects. Particular emphasis should be placed on moderators that put differential emphasis on one engagement layer or another—for example, independent and interdependent selves (Ng & Houston, 2006; Swaminathan et al., 2007) or self-brand engagement (Sprott et al., 2009).

Consumer neuroscience

Most of the research that I am suggesting for testing the model can be done using standard research techniques such as experiments, surveys and causal modeling. However, because we proposed that the five processes of the model might not be stepwise but widely distributed, newly emerging consumer insight techniques may be useful as well. For example, consumer neuroscience addresses consumer relevant problems using methods originally used for brain research (Hubert & Kenning, 2008). Brain imaging techniques examine activations as they occur spontaneously and simultaneously throughout the entire brain and may thus help to test and refine the proposed model (Shiv et al., 2005; Yoon, Gutchess, Feinberg, & Polk, 2006).

Consumer neuroscience on brands has provided some preliminary evidence that is consistent with the model. Consider the well known “Coke-Pepsi” brain study by McClure et al. (2004). While observing the typical “Pepsi challenge” effect that consumers prefer Pepsi in a blind test over Coke, the authors also observed stronger activations in brain regions associated with emotions when participants were shown a Coke logo than a Pepsi logo. Importantly, the effect could be reversed with patients whose ventromedial prefrontal cortex, an emotion area, was damaged: they preferred Pepsi to Coke both in a blind taste test and in a test that featured brand information (Koenigs & Tranel, 2008). These and other results suggest that brand choice can be driven by sensory information in some contexts, but also by experienced emotions (Esch et al., 2012-this issue). Moreover, results from a series of brain scanning studies suggest that brands may be processed differently for extracting information for functional decision-making, personal meaning and inter-personal relevance. For example, the prefrontal cortex seems to be differentially engaged for value versus prestige brands; moreover, the medial prefrontal cortex and precuneus seem to be involved in the processing of luxury brands (associated with self-centered cognitions) whereas the left superior frontal gyrus and anterior cingulate cortex are activated for value brands that are more likely to follow rather objective, functional choice (Schaefer, 2009). Finally, abstract, semantic brand concepts can be neurologically distinguished from affective, self-relevant brand concepts (Ratnayake, Broderick, & Mitchell, 2010).

In future consumer-neuroscience research, the model should be tested more comprehensively and rigorously, e.g., by manipulating levels of engagement, rather than measuring them through brands or category types. This could be done by priming the levels of engagement and giving individuals simple tasks related to the five processes of identifying, experiencing, integrating, signifying and connecting, while measuring activations in brain regions.

Conclusion

While research in the branding area has usually been narrow and not integrative, some have been waiting for a comprehensive consumer-psychology model of brands—a “blueprint of brand knowledge, as comprehensive while also as parsimonious as possible, that would provide the necessary depth and breadth of understanding of consumer behavior and marketing activity” (Keller, 2003). I feel that the model that I presented here has achieved this kind of comprehensive, yet parsimonious, blueprint of brands from a consumer psychology perspective. The model, developed primarily for an academic audience, summarizes and integrates existing constructs and findings and intends to stimulate more systematic future research. By drawing the attention of practitioners to the key psychological factors underlying brand effects, the model may also provide a bridge from theory to business practice and inspire managers to create functionally useful, psychologically meaningful and culturally relevant brands.

References

- Aaker, D. (1991). *Managing brand equity: Capitalizing on the value of a brand name*. New York: The Free Press.
- Aaker, D. (1996). *Building strong brands*. New York: The Free Press.
- Aaker, J. L. (1997). Dimensions of brand personality. *Journal of Marketing Research*, 34, 447–356.
- Aaker, J. L. (1999). The malleable self: The role of self-expression in persuasion. *Journal of Marketing Research*, 36, 45–57.
- Aaker, D. (2004). *Brand portfolio strategy*. New York: The Free Press.
- Aaker, J., Benet-Martinez, V., & Garolera, J. (2001). Consumption symbols as carriers of culture: A study of Japanese and Spanish brand personality constructs. *Journal of Personality and Social Psychology*, 81, 492–508.
- Aaker, J. L., Fournier, S., & Brasel, S. A. (2004). When good brands do bad. *Journal of Consumer Research*, 31, 1–16.
- Aaker, D. A., & Keller, K. L. (1990). Consumer evaluations of brand extensions. *Journal of Marketing*, 54, 27–41.
- Adaval, R. (2003). How good gets better and bad gets worse: Understanding the impact of affect on evaluations of known brands. *The Journal of Consumer Research*, 30, 352–367.
- Agarwal, M. K., & Rao, V. R. (1996). An empirical comparison of consumer-based measures of brand equity. *Marketing Letters*, 3, 237–247.
- Aggarwal, P. (2004). The effects of brand relationship norms on consumer attitudes and behavior. *The Journal of Consumer Research*, 30, 87–101.
- Aggarwal, P., & Law, S. (2005). Role of relationship norms in processing brand information. *Journal of Consumer Research*, 32, 453–464.
- Aggarwal, P., & McGill, A. L. (2007). Is this car smiling at me? Schema congruity as basis for evaluating anthropomorphized products. *Journal of Consumer Research*, 34, 468–479.
- Akaoui, J. (2007). Brand experience on the pitch: How the sponsors fared in the World Cup. *Journal of Advertising Research*, 47, 147–157.
- Alba, J. W., & Chattopadhyay, A. (1985). Effects of context and part-category cues on recall of competing brands. *Journal of Marketing Research*, 22, 340–349.
- Alba, J. W., Hutchinson, W. J., & Lynch, J. (1991). Memory and decision making. In Thomas S. Robertson, & Harold H. Kassarijian (Eds.), *Handbook of Consumer Behavior* (pp. 1–49). NJ: Englewood Cliffs.
- Algesheimer, R., Dholakia, U. M., & Herrmann, A. (2005). The social influence of brand community: Evidence from European car clubs. *Journal of Marketing*, 69, 19–34.
- Algom, D., & Cain, W. S. (1991). Remembered odors and mental mixtures: Tapping reservoirs of olfactory knowledge. *Journal of Experimental Psychology. Human Perception and Performance*, 17, 1104–1119.

- Anderson, N. H. (1981). *Foundations of information integration theory*. San Diego, CA: Academic Press.
- Andrade, E. B., & Cohen, J. B. (2007). On the consumption of negative feelings. *Journal of Consumer Research*, 34, 283–300.
- Arnould, E. J. (2005). Animating the big middle. *Journal of Retailing*, 81(2), 89–96.
- Bagozzi, R. P., & Dholakia, U. M. (2006). Antecedents and purchase consequences of customer participation in small group brand communities. *International Journal of Research in Marketing*, 23, 45–61.
- Barone, M. J. (2005). The interactive effects of mood and involvement on brand extension evaluations. *Journal of Consumer Psychology*, 15, 263–270.
- Barone, M. J., Miniard, P. W., & Romeo, J. B. (2000). The influence of positive mood on brand extension evaluations. *The Journal of Consumer Research*, 26, 340–357.
- Bayus, B. L. (2010). **Crowdsourcing and individual creativity over time: The detrimental effects of past success. Working Paper Series.**
- Bottomley, P. A., & Holden, S. J. S. (2001). Do we really know how consumers evaluate brand extensions? Empirical generalizations based on secondary analysis of eight studies. *Journal of Marketing Research*, 38, 494–500.
- Boush, D. M., & Loken, B. (1991). A process-tracing study of brand extension evaluation. *Journal of Marketing Research*, 28, 16–28.
- Bowlby, J. (1979). *The making and breaking of affectional bonds*. London: Tavistock.
- Brakus, J. Josko, Schmitt, B. H., & Zarantonello, L. (2009). Brand experience: What is it? How is it measured? Does it affect loyalty? *Journal of Marketing*, 73, 52–68.
- Bridges, S., Keller, K. L., & Sood, S. (2000). Communication strategies for brand extensions: Enhancing perceived fit by establishing explanatory links. *Journal of Advertising*, 29, 1–11.
- Broniarczyk, S. M., & Alba, J. W. (1994). The importance of the brand in brand extension. *Journal of Marketing Research*, 31, 214–228.
- Calvert, G. A., Spence, C., & Stein, B. E. (2004). *The handbook of multisensory processes*. Cambridge, MA: MIT Press.
- Caprara, G. V., Barbaranelli, C., & Guido, G. (2001). Brand personality: How to make the metaphor fit? *Journal of Economic Psychology*, 22, 377–395.
- Carroll, B. A., & Ahuvia, A. C. (2006). Some antecedents and outcomes of brand love. *Marketing Letters*, 17, 79–89.
- Chaplin, L. N., & Roedder John, D. (2005). The development of self-brand connections in children and adolescents. *Journal of Consumer Research*, 32, 119–129.
- Chaudhuri, A., & Holbrook, M. B. (2001). The chain of effects from brand trust and brand affect to brand performance: The Role of Brand Loyalty. *Journal of Marketing*, 65, 81–93.
- Delgado-B, E., Munuera-Aleman, J. L., & Yagoe-Guillin, M. J. (2003). Development and validation of a brand trust scale. *International Journal of Market Research*, 45, 335–353.
- Dimofte, C. V., & Yalch, R. F. (2011). The mere association effect and brand evaluations. *Journal of Consumer Psychology*, 21, 24–37.
- Erdem, T., & Swait, J. (1998). Brand equity as a signaling phenomenon. *Journal of Consumer Psychology*, 7, 131–157.
- Escalas, J. E. (2004). Narrative processing: Building consumer connections to brands. *Journal of Consumer Psychology*, 14, 168–179.
- Esch, F.-R., Möll, T., Schmitt, B., Elger, C., Neuhaus, C., & Weber, B. (2012). Brands on the brain: Do consumers use declarative information or experienced emotions to evaluate brands? *Journal of Consumer Psychology*, 22, 75–85 this issue.
- Farquhar, P. H., & Herr, P. M. (1993). The dual structure of brand associations. In David A. Aaker, & Alexander L. Biel (Eds.), *Brand equity & advertising: Advertising's role in building strong brands* (pp. 263–279). Hillsdale, NJ: Laurence Erlbaum Associates.
- Fedorikhin, A., Park, C. W., & Thomson, M. (2008). Beyond fit and attitude: The effect of emotional attachment on consumer responses to brand extensions. *Journal of Consumer Psychology*, 18(4), 281–291.
- Fournier, S. (1998). Consumers and their brands: Developing relationship theory in consumer research. *Journal of Consumer Research*, 24, 343–373.
- Fournier, S., & Yao, J. L. (1997). Reviving brand loyalty: A reconceptualization within the framework of consumer-brand relationships. *International Journal of Research in Marketing*, 14(5), 451–472.
- Gawronski, B., & Bodenhausen, G. V. (2006). Associative and propositional processes in evaluation: An integrative review of implicit and explicit attitude change. *Psychological Bulletin*, 132, 692–731.
- Geuens, M., Weijters, B., & De Wulf, K. (2009). A new measure of brand personality. *International Journal of Research in Marketing*, 26, 97–107.
- Geylani, T., Ter Hofstede, F., & Inman, J. (2005). *Image reinforcement or impairment: The effects of co-branding on attribute uncertainty*. McCombs Working Paper No. MKT-01-05.
- Gobé, M. (2001). *Emotional branding: The new paradigm for connecting brands to people*. Allworth Press.
- Goldstein, N. J., & Cialdini, R. B. (2007). The spyglass self: A model of vicarious self-perception. *Journal of Personality and Social Psychology*, 92(3), 402–417.
- Graeff, T. R. (1997). Consumption situations and the effects of brand image on consumers' brand evaluations. *Psychology and Marketing*, 14, 49–70.
- Grewal, D., Kavanoor, S., Fern, E. F., Costley, C., & Barnes, J. (1997). Comparative versus noncomparative advertising: A meta-analysis. *The Journal of Marketing*, 61, 1–15.
- Grossman, R. P., & Till, B. D. (1998). The persistence of classically conditioned brand attitudes. *Journal of Advertising*, 27, 23–31.
- Hazan, C., & Shaver, P. R. (1994). Attachment as an organizational framework for research on close relationships. *Psychological Inquiry*, 5, 1–22.
- Herr, P. M., Farquhar, P. H., & Fazio, R. H. (1996). Impact of dominance and relatedness on brand extensions. *Journal of Consumer Psychology*, 5, 135–159.
- Hoffman, D., & Fodor, M. (2010). Can you measure the ROI of your social media marketing? *MIT Sloan Management Review*, 52(1).
- Holt, D. B. (2004). *How brands become icons: The principles of cultural branding*. Cambridge: Harvard Business Press.
- Hoyer, W. D., & Brown, S. P. (1990). Effects of brand awareness on choice for a common, repeat-purchase product. *The Journal of Consumer Research*, 17, 141–148.
- Hubbard, T. L. (1994). Memory psychophysics. *Psychological Research*, 56, 237–250.
- Hubert, M., & Kenning, P. (2008). A current overview of consumer neuroscience. *Journal of Consumer Behaviour*, 7, 272–292.
- Janiszewski, C., & van Osselaer, S. M. J. (2000). A connectionist model of brand-quality associations. *Journal of Marketing Research*, 37, 331–350.
- Keller, K. L. (1987). Memory factors in advertising: The effect of advertising retrieval cues on brand evaluations. *The Journal of Consumer Research*, 14, 316–333.
- Keller, K. L. (1993). Conceptualizing, measuring, and managing consumer-based brand equity. *Journal of Marketing*, 57, 1–22.
- Keller, K. L. (2002). Branding and brand equity. In Bart Weitz, & Robin Wensley (Eds.), *Handbook of Marketing* (pp. 151–178). London: Sage Publications.
- Keller, K. L. (2003). Brand synthesis: The multidimensionality of brand knowledge. *Journal of Consumer Research*, 29, 595–600.
- Keller, K. L., & Aaker, D. A. (1992). The effects of sequential introduction of brand extensions. *Journal of Marketing Research*, 29, 35–50.
- Keller, K. L., & Lehman, D. R. (2006). Brands and branding: Research findings and future priorities. *Marketing Science*, 25, 740–759.
- Kirmani, A. (2009). The self and the brand. *Journal of Consumer of Business*, 19, 271–275.
- Koenigs, M., & Tranel, D. (2008). Prefrontal cortex damage abolishes brand-cued changes in cola preference. *Social Cognitive and Affective Neuroscience*, 3, 1–6.
- Kreuzbauer, R., & Malter, A. J. (2005). Embodied cognition and new product design: Changing product form to influence brand categorization. *Journal of Product Innovation Management*, 22, 165–176.
- Leclerc, F., Schmitt, B. H., & Dubé, L. (1994). Foreign branding and its effects on product perceptions and attitudes. *Journal of Marketing Research*, 31, 264–270.
- Lowrey, T. M., Shrum, L. J., & Dubitsky, T. M. (2003). The Relation between brand-name linguistic characteristics and brand-name memory. *Journal of Advertising*, 32, 7–17.
- Lynch, J. G., & Srull, T. K. (1982). Memory and attentional factors in consumer choice: Concepts and research methods. *Journal of Consumer Research*, 9, 18–37.

- MacKenzie, S. B., Lutz, R. J., & Belch, G. E. (1986). The role of attitude toward the ad as a mediator of advertising effectiveness: A test of competing explanations. *Journal of Marketing Research*, 23, 130–143.
- Maheswaran, D., Mackie, D. M., & Chaiken, S. (1992). Brand name as a heuristic cue: The effects of task importance and expectancy confirmation on consumer judgments. *Journal of Consumer Psychology*, 1, 317–336.
- Maoz, E., & Tybout, A. M. (2002). The moderating role of involvement and differentiation in the evaluation of brand extensions. *Journal of Consumer Psychology*, 119–131.
- Markus, H. (1977). Self-schemata and processing information about the self. *Journal of Personality and Social Psychology*, 35, 63–78.
- Markus, H. (1983). Self-knowledge: An expanded view. *Journal of Personality*, 51, 543–565.
- Markus, H., & Sentsis, K. (1982). The self in social information processing. In J. Suls (Ed.), *Psychological perspectives on the self* (pp. 41–70). Hillsdale, NJ: Edbaum.
- McAlexander, J. H., Schouten, J. W., & Koenig, H. J. (2002). Building brand community. *Journal of Marketing*, 66, 38–54.
- McClure, S. M., Li, J., Tomlin, D., Cypert, K. S., Montague, L. M., & Montague, P. R. (2004). Neural correlates of behavioral preference for culturally familiar drinks. *Neuron*, 44, 379–387.
- McCrae, R. R., & Costa, P. T., Jr. (1997). Personality trait structure as a human universal. *American Psychologist*, 52, 509–516.
- Mick, D. G. (1986). Consumer research and semiotics: Exploring the morphology of signs, symbols and significance. *Journal of Consumer Research*, 13, 196–214.
- Milberg, S. J., Park, C. W., & McCarthy, M. S. (1997). Managing negative feedback effects associated with brand extensions: The impact of alternative branding strategies. *Journal of Consumer Psychology*, 6, 119–140.
- Milberg, S. J., Sinn, F., & Goodstein, R. C. (2010). Consumer reactions to brand extensions in a competitive context: Does fit still matter? *Journal of Consumer Research*, 37, 543–553.
- Mitchell, A. A., & Olson, J. C. (1981). Are product attribute beliefs the only mediator of advertising effects on brand attitude? *Journal of Marketing Research*, 18, 318–332.
- Monga, A. B., & John, D. R. (2010). What makes brands elastic? The influence of brand concept and styles of thinking on brand extension evaluation. *Journal of Marketing*, 74, 80–92.
- Morrin, M. (1999). The impact of brand extensions on parent brand memory structures and retrieval processes. *Journal of Marketing Research*, 36, 517–525.
- Morrin, M., & Ratneswar, S. (2003). Does it make sense to use scents to enhance brand memory. *Journal of Marketing Research*, 40, 10–25.
- Moscovici, S., & Markova, I. (1998). Presenting social representations: A conversation. *Culture and Psychology*, 4, 371–410.
- Muniz, A. M., & O'Guinn, T. C. (2001). Brand community. *The Journal of Consumer Research*, 27, 412–432.
- Nedungadi, P. (1990). Recall and consumer consideration sets: Influencing choice without altering brand evaluations. *Journal of Consumer Research*, 17, 263–276.
- Ng, S., & Houston, M. J. (2006). Exemplars or beliefs? The impact of self-view on the nature and relative influence of brand associations. *Journal of Consumer Research*, 32, 519–529.
- Park, C. W., Jaworski, B. J., & MacInnis, D. J. (1986). Strategic brand concept-image management. *Journal of Marketing*, 50, 135–145.
- Park, C. W., & MacInnis, D. J. (2006). What's in and what's out: Questions on the boundaries of the attitude construct. *Journal of Consumer Research*, 33, 16–18.
- Park, C. W., MacInnis, D. J., Priester, J., Eisingerich, A. B., & Iacobucci, D. (2010). Brand attachment and brand attitude strength: Conceptual and empirical differentiation of two critical brand equity drivers. *Journal of Marketing*, 74, 1–17.
- Park, C. S., & Srinivasan, V. (1994). A survey-based method for measuring and understanding brand equity and its extendibility. *Journal of Marketing Research*, 31, 271–288.
- Pham, M. T. (2004). Ideals and oughts and the reliance on affect versus substance in persuasion. *The Journal of Consumer Research*, 30, 503–518.
- Pine, J., & Gilmore, J. (1999). *The experience economy*. Boston: Harvard Business School Press.
- Pullig, C., Simmons, C. J., & Netemeyer, R. G. (2006). Brand dilution: When do new brands hurt existing brands? *Journal of Marketing*, 70, 52–66.
- Ramanathan, S., & Williams, P. (2007). Immediate and delayed emotional consequences of indulgence: The moderating influence of personality type on mixed emotions. *Journal of Consumer Research*, 34, 212–223.
- Ratnayake, N., Broderick, A. J., & Mitchell, R. L. C. (2010). A neurocognitive approach to brand memory. *Journal of Marketing Management*, 26, 1295–1318.
- Rindfleisch, A., Burroughs, J., & Wong, N. (2009). The safety of objects: materialism, existential insecurity and brand connection. *Journal of Consumer Research*, 36, 1–16.
- Roedder John, D. R., Loken, B., Kim, K., & Monga, A. B. (2006). Brand concept maps: A methodology for identifying brand association networks. *Journal of Marketing Research*, 43, 549–563.
- Schaefer, M. (2009). Neuroeconomics: In search of the neural representation of brand. *Progress in Brain Research*, 178, 241–252.
- Schmitt, B. (1999). *Experiential marketing: How to get customers to sense, feel, think, act, and relate to your company and brands*. New York: The Free Press.
- Schmitt, B. (2003). *Customer experience management: A revolutionary approach to connecting with your customers*. Hoboken, New Jersey: John Wiley & Sons.
- Schmitt, B. H., Tavassoli, N. T., & Millard, R. T. (1993). Memory for print ads: Understanding relations among brand name, copy, and picture. *Journal of Consumer Psychology*, 2, 55–81.
- Schwarz, N. (1990). Feelings as information: Informational and motivational functions of affective states. In R. M. Sorrentino, & E. T. Higgins (Eds.), *Handbook of motivation and cognition: Foundations of social behavior*, Vol. 2. (pp. 527–561) New York: Guilford Press.
- Sen, S. (1999). The effects of brand name suggestiveness and decision goal on the development of brand knowledge. *Journal of Consumer Psychology*, 8, 431–455.
- Shapiro, S., & Spence, M. T. (2002). Factors affecting encoding, retrieval, and alignment of sensory attributes in a memory based brand choice task. *Journal of Consumer Research*, 28, 603–617.
- Sheinin, D. A., & Schmitt, B. H. (1994). Extending brands with new product concepts: The role of category attribute congruity, brand affect, and brand breadth. *Journal of Business Research*, 31, 1–10.
- Shiv, B., Bechara, A., Levin, I., Alba, J., Bettman, J., Dubé, L., et al. (2005). Decision neuroscience. *Marketing Letters*, 16, 375–386.
- Simonin, B. L., & Ruth, J. A. (1998). Is a company known by the company it keeps? Assessing the spillover effects of brand alliances on consumer brand attitudes. *Journal of Marketing Research*, 35, 30–42.
- Smith, C. A., & Ellsworth, P. C. (1985). Patterns of cognitive appraisal in emotion. *Journal of Personality and Social Psychology*, 48, 813–838.
- Sood, S., & Dreze, X. (2006). Brand extensions of experiential goods: Movie sequel evaluations. *Journal of Consumer Research*, 33, 352–360.
- Spence, C. (2010). Crossmodal attention. *Scholarpedia*, 5, 6309.
- Sprott, D. E., Czellar, S., & Spangenberg, E. (2009). The importance of a general measure of brand engagement on market behavior: Development and validation of a scale. *Journal of Marketing Research*, 46, 92–104.
- Swaminathan, V., Page, K. L., & Gürhan-Canli, Z. (2007). My' brand or 'our' brand: The effects of brand relationship dimensions and self-construal on brand evaluations. *Journal of Consumer Research*, 34(August), 248–259.
- Tavassoli, N. T. (2001). Color memory and evaluations for alphabetical and logographic brand names. *Journal of Experimental Psychology. Applied*, 7, 104–111.
- Thomson, M., MacInnis, D. J., & Park, W. C. (2005). The ties that bind: Measuring the strength of consumers' emotional attachments to brands. *Journal of Consumer Psychology*, 15(1), 77–91.
- Trinke, S. J., & Bartholomew, K. (1997). Hierarchies of attachment relationships in young adulthood. *Journal of Social and Personal Relationships*, 15, 603–625.
- Weiss, R. S. (1988). Loss and recovery. *Journal of Social Issues*, 44, 37–52.
- Woodside, A. G., Sood, S., & Miller, K. E. (2008). When consumers and brands talk: Storytelling theory and research in psychology and marketing. *Psychology and Marketing*, 25, 97–145.
- Yoon, C., Gutches, A. H., Feinberg, F., & Polk, T. A. (2006). A functional magnetic resonance imaging study of neural dissociations between brand and person judgments. *Journal of Consumer Research*, 33, 31–40.

- Yorkston, E., & Menon, G. (2004). A sound idea: Phonetic effects of brand names on consumer judgments. *The Journal of Consumer Research*, 31, 43–51.
- Zampini, M., & Spence, C. (2004). The role of auditory cues in modulating the perceived crispness and staleness of potato chips. *Journal of Sensory Studies*, 19, 347–363.
- Zeithaml, V. A. (1988). Consumer perceptions of price, quality, and value: A means-end model and synthesis of evidence. *Journal of Marketing*, 52, 2–22.
- Zhang, S., & Schmitt, B. H. (2001). Creating local brands in multilingual international marketing. *Journal of Marketing Research*, 38, 313–325.